Identify Horses or Humans with TensorFlow and Vertex AI

experimentLabschedule1 hour 30 minutesuniversal_currency_alt5 Creditsshow_chartIntermediate

infoThis lab may incorporate AI tools to support your learning.

GSP634

Overview

In other labs the Fashion MNIST dataset is used to train an image classifier. In this case you had images that were 28x28 where the subject was centered. In this lab you'll take this to the next level, training to recognize features in an image where the subject can be *anywhere* in the image!

You'll do this by building a horses-or-humans classifier that will tell you if a given image contains a horse or a human, where the network is trained to recognize features that determine which is which.

Objectives

In this lab, you will learn how to:

- Look at the starting model
- Add convolutions, gather the data, and define the model
- Compile and train the model
- Visualize the Convolutions and pooling

Prerequisites

Although this is a self-standing lab, to maximize your learning, consider taking the lab, Classify Images with TensorFlow Convolutional Neural Networks.

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources will be made available to you.

This hands-on lab lets you do the lab activities yourself in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials that you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito or private browser window to run this lab. This prevents any conflicts between your personal account and the Student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab---remember, once you start, you cannot pause a lab.

Note: If you already have your own personal Google Cloud account or project, do not use it for this lab to avoid extra charges to your account.

How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:

    - The **Open Google Cloud console** button
    - Time remaining
    - The temporary credentials that you must use for this lab
    - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

    The lab spins up resources, and then opens another tab that shows the **Sign in** page.

    Tip: Arrange the tabs in separate windows, side-by-side.

    Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

    ```
    "Username"
    ```

Copied!

content_copy

You can also find the **Username** in the **Lab Details** panel.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

   ```
   "Password"
   ```

   Copied!

   content_copy

   You can also find the **Password** in the **Lab Details** panel.

6. Click **Next**.

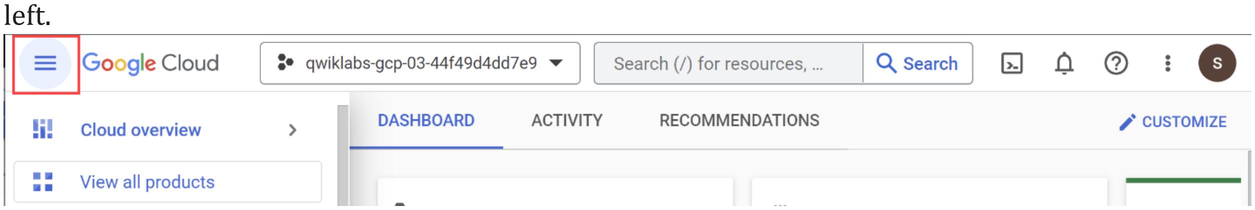
   Important: You must use the credentials the lab provides you. Do not use your Google Cloud account credentials.**Note:** Using your own Google Cloud account for this lab may incur extra charges.

7. Click through the subsequent pages:

   - Accept the terms and conditions.
   - Do not add recovery options or two-factor authentication (because this is a temporary account).
   - Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.

Note: To view a menu with a list of Google Cloud products and services, click the **Navigation menu** at the top-left.



Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell** at the top of the Google Cloud console.

When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `PROJECT_ID`. The output contains a line that declares the **Project_ID** for this session:

```
Your Cloud Platform project in this session is set to "PROJECT_ID"
```

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

2. (Optional) You can list the active account name with this command:

```
gcloud auth list
```

Copied!

content_copy

3. Click **Authorize**.

Output:

```
ACTIVE: *
ACCOUNT: "ACCOUNT"

To set the active account, run:
    $ gcloud config set account `ACCOUNT`
```

4. (Optional) You can list the project ID with this command:

```
gcloud config list project
```

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Output:

```
[core]
project = "PROJECT_ID"
```

Note: For full documentation of `gcloud`, in Google Cloud, refer to [the gcloud CLI overview guide](#).

Task 1. Open the notebook in Vertex AI Workbench

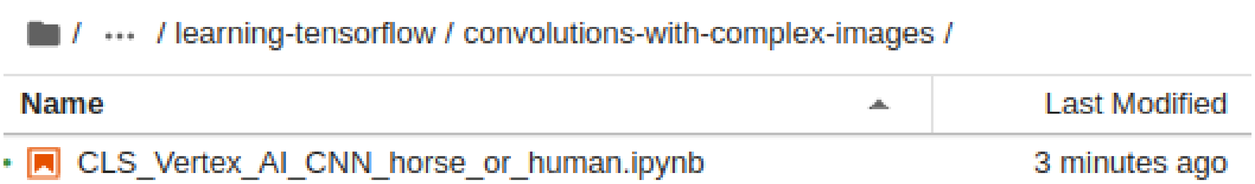
1. In the Google Cloud console, on the **Navigation menu** (), click **Vertex AI > Workbench**.

2. Find the `Workbench instance name` instance and click on the **Open JupyterLab** button.

The JupyterLab interface for your Workbench instance opens in a new browser tab.

Task 2. Run the lab notebook

1. In the left hand menu, modify `Notebook name` to include your region in cell 6. You can get it in the left panel of the lab instructions.



2. Continue the lab in the notebook, and run each cell by clicking the **Run** ( ) icon at the top of the screen. Alternatively, you can execute the code in a cell with **SHIFT + ENTER**.

Read the narrative and make sure you understand what's happening in each cell.

In order to view the status of training and deployment on Vertex AI, you can follow the instructions in the notebook containing illustrations.

Check your progress on notebook

Create a Cloud Storage Bucket

Note: While creating the bucket change the `REGION` that has been assigned to you in the student resource panel.

Click *Check my progress* to verify whether the bucket is created.

Create a Cloud Storage Bucket

Check my progress

Train the Model on Vertex AI

Click *Check my progress* to verify the objective.

Train the Model on Vertex AI

Check my progress

Copy the Batch Prediction Results

Click *Check my progress* to verify the objective.

Copy the Batch Prediction Results

Check my progress

Congratulations!

In this lab, you used convolution to identify features in an image where the subject could be anywhere in the image!

Next steps / learn more

- Check out the Tensorflow documentation on [ImageDataGenerator]

- Learn more about the at Keras DataGenerator.

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. Our classes include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. Certifications help you validate and prove your skill and expertise in Google Cloud technologies.
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